LAB 3 - COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

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System: Self-Service Coffee Kiosk

Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Reason 1 - Clear Separation of Responsibilities

Layered Architecture separates the kiosk into Presentation, Business and Data layers. The User Interface Component handles touch-screen interaction, while the Order Manager, Payment Service and Receipt Printer handle business operations. The Database Component stores and retrieves menu, pricing and order information. This separation makes the system easier to understand, maintain and modify.

Reason 2 - Appropriate for a Single Kiosk

The coffee kiosk is a relatively small application with a limited number of closely related components. A Microservices architecture would introduce unnecessary operational and network complexity, while Client-Server architecture could introduce a central-server dependency. Layered Architecture therefore provides a simpler and more appropriate design.

Security Advantage

Payment processing is isolated within the Payment Service Component. Credit-card operations can therefore be separated from the User Interface and Database components, reducing unnecessary exposure of sensitive payment information and allowing payment-related validation and security controls to be concentrated in one component.

Performance Benefit

The architecture provides a direct and predictable flow from the User Interface to the Order Manager and required services. Because the kiosk uses a compact layered structure, unnecessary distributed network communication is avoided. This supports fast interactions and helps satisfy the requirement that an order be completed in under 60 seconds.